ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | 979-8-3503-6874-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICASSP49660.2025.10887772

RadDet: A Wideband Dataset for Real-Time Radar Spectrum Detection

Zi Huang*†, Simon Denman*, Akila Pemasiri*, Terrence Martin*†, Clinton Fookes*
*School of Electrical Engineering & Robotics, Queensland University of Technology, Brisbane, Australia
†Revolution Aerospace, Brisbane, Australia
Email: {z37.huang, s.denman, a.thondilege, tl.martin, c.fookes}@qut.edu.au

***Abstract*—Real-time detection of radar signals in a wideband radio frequency spectrum is a critical situational assessment function in electronic warfare. Compute-efficient detection models have shown great promise in recent years, providing an opportunity to tackle the spectrum detection problem. However, progress in radar spectrum detection is limited by the scarcity of publicly available wideband radar signal datasets accompanied by corresponding annotations. To address this challenge, we introduce a novel and challenging dataset for radar detection (RadDet), comprising a large corpus of radar signals occupying a wideband spectrum across diverse radar density environments and signal-to-noise ratio (SNR) settings. RadDet contains 40,000 frames, each generated from 1 million in-phase and quadrature (I/Q) samples across a 500 MHz frequency band. RadDet includes 11 classes of radar signals across 6 different SNR settings, 2 radar density environments, and 3 different time-frequency resolutions, with corresponding time-frequency and class annotations. We evaluate the performance of state-of-the-art real-time detection models on RadDet and a modified radar classification dataset from NIST (NIST-CBRS) to establish a novel benchmark for wideband radar spectrum detection.**

***Index Terms*—spectrum sensing, radar signal recognition, real-time detection, wideband radar dataset**

I. Introduction

Real-time detection and classification of complex radar waveforms in degraded signal-to-noise ratio (SNR) environments are fundamental requirements in cognitive electronic warfare (CEW) [1], [2]. These functions must be performed at the front-end of a CEW system to detect and localise potential threats within the radar spectrum. Identifying the dwell time and bandwidth of a radar signal is a critical step in attaining situational awareness of the radar environment, enabling the development of effective radio frequency (RF) countermeasures [1]. Furthermore, in a highly contested electronic warfare (EW) setting with numerous potential RF threats, these tasks must be executed in real-time across a large frequency band to ensure wide spectrum coverage and timely deployment of RF countermeasures [2]. However, the scarcity of publicly available wideband radar signal datasets has limited progress in real-time radar spectrum detection.

In this work, we present a large-scale dataset to support the study of radar spectrum detection in CEW. We focus on the non-cooperative detection of synthetic radar threats across a wide frequency band with overlapping frequency content, and in degraded SNR settings. We then investigate the application of compute-efficient detection models to perform classification and localisation of time-frequency occupancy, and explore the impact of time-frequency resolution on detection performance and speed. To encourage future research in CEW, we present several variants of the dataset that explore different synthetic radar density environments. The main contributions of this paper are as follows: (i) we introduce a large-scale open-source dataset[1] for wideband radar spectrum detection; and (ii) we establish of a new benchmark for wideband radar spectrum detection across different radar density environments and provide empirical results, analysing the impact of time-frequency resolution on detection performance.

II. Related Work

Identifying an RF waveform in the presence of background noise through automatic signal recognition (ASR) has been an active research area over the past decade [3]–[10]. Recent advancements in ASR have employed various approaches, including the short-time Fourier transform (STFT) [11], along with task-specific representations such as the wavelet transform [12], Wigner-Ville distribution [13], and Choi-Williams distribution [14], to generate time-frequency representations from raw signals for subsequent analysis. The STFT is often preferred due to its consistent time-frequency resolution, while other transforms may introduce undesirable artefacts [15] in the time-frequency domain and require careful design to be tailored for specific tasks. Other approaches used statistical priors [16] and feature engineering [8], [9] for feature extraction, however they do not cater for non-cooperative and covert spectrum users encountered in an EW setting. Recent advances in deep convolutional neural networks (CNN) and transformer-based architectures [17], [18] have shown promise for ASR, with applications in automatic modulation classification (AMC) [7], [18], [19], radar pulse activity segmentation [20], and spectrum sensing [16], [21], [22]. However, research into radar spectrum detection is limited by the scarcity of publicly available radar detection datasets [23].

The challenge of ASR in EW lies in detecting and classifying low probability of intercept (LPI) [24], [25] radar signals across a wide frequency band [1], [26], requiring a careful trade-off between spectrum resolution and coverage. Furthermore, localising the occupancy of radar threats across

[1]Our radar spectrum detection dataset and its variants can be accessed at: https://github.com/abcxyzi/RadDet

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI. Downloaded on August 06,2026 at 15:39:50 UTC from IEEE Xplore. Restrictions apply.

a wide spectrum within an EW context is particularly difficult due to the need for near real-time performance [2]. While recent advancements in ASR have led to the development of several RF datasets, such as RadioML [7], RadarComms [19], [27], DeepRadar [25], and RadChar [18], they are constrained by short I/Q sequences, narrow frequency bands, and lack of annotations for time-frequency localisation. RadSeg [20] is a large-scale dataset that explores the segmentation of interleaved radar pulse activities while considering long I/Q sequences. This dataset focuses on 5 radar classes, however it provides only temporal annotations. Another dataset is the RF Dataset of Incumbent Radar Systems in the 3.5 GHz CBRS Band (NIST-CBRS), which was developed by the National Institute of Standards and Technology (NIST) [28], [29]. It provides $40,000$ long-duration waveforms containing 5 radar classes, sampled at 10 MHz. However, this dataset is limited to the detection of a single radar instance across a narrow SNR range. Other studies [11], [30], [31] have investigated the near real-time detection of multiple radar and telecommunications waveforms using CNNs, however the RF datasets used in these works are not publicly accessible.

III. Radar Datasets

We present two large-scale datasets for radar spectrum detection. The first dataset extends the existing work by NIST [29], providing a new baseline for real-time radar detection. The second dataset, developed in this work, intends to challenge contemporary detection algorithms by exploring a wider radar spectrum, a broader range of radar waveforms, varied radar density environments, and lower SNR settings.

A. NIST CBRS 3.5 GHz Dataset

NIST-CBRS [28] is a synthetic radar dataset comprising 5 radar classes, including two pulsed radars and three linear frequency-modulated (LFM) radars, sampled at 10 MHz with SNR levels ranging from 10 to 20 dB. The dataset includes $40,000$ baseband I/Q signals, each $800,000$ samples in length. We transform the dataset into a spectrum detection dataset, providing time-frequency annotations as shown in Fig. 1. The dimensions of the time-frequency bounding box (bbox) for each signal are calculated by

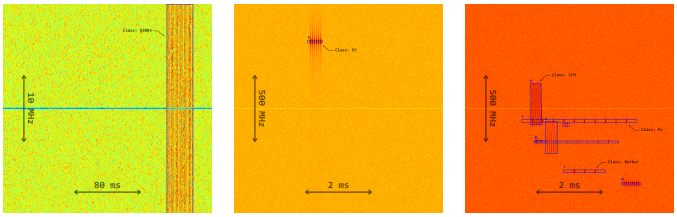
$$t_{\text{bbox}} \in \left[t_s, t_s + \frac{N}{f_{\text{prf}}} \right], \tag{1}$$

$$f_{\text{bbox}} \in \left[f_{\text{c}} - \frac{B}{2}, f_{\text{c}} + \frac{B}{2} \right], \tag{2}$$

where t_{bbox} denotes the bbox width (i.e., temporal occupancy), f_{bbox} denotes the bbox height (i.e., frequency occupancy), N denotes the sample length, f_{c} denotes the centre frequency, f_{prf} denotes the pulse repetition frequency (PRF), and B denotes the signal bandwidth which is estimated by

$$B = \begin{cases} B_{\text{chirp}} & \text{if frequency-modulated,} \\ t_{\text{pw}}^{-1} + \epsilon & \text{if pulsed or phase-coded.} \end{cases} \tag{3}$$

For LFM radars (i.e., Q3N classes), B is given by the chirp bandwidth B_{chirp}. For pulsed radars (i.e., P0N classes), the reciprocal of the signal's pulse width t_{pw} is used to estimate B. As a clean signal was not provided in the original dataset to precisely compute B for pulsed radars, a constant buffer ϵ estimated by $0.5 \times t_{\text{pw}}^{-1}$ is added to account for any measurement uncertainty.



(a) NIST-CBRS (b) RadDet-1T (c) RadDet-9T

Fig. 1. Radar detection frames with time-frequency annotations (bounding boxes) shown at 10 dB SNR.

TABLE I
Summary of RadDet Signal Parameters

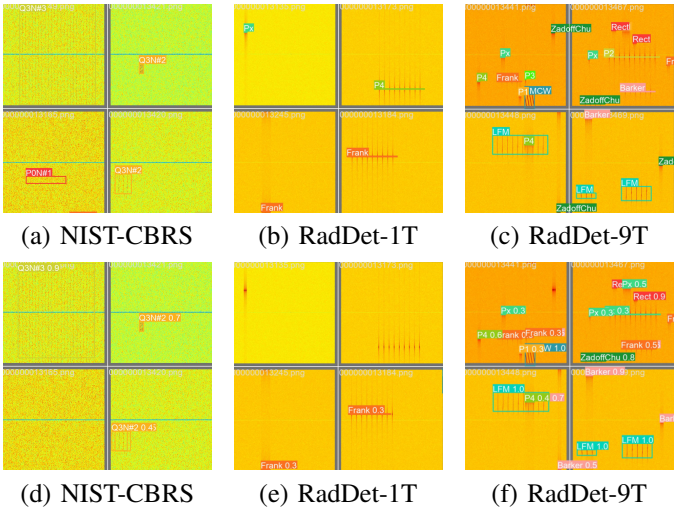
Signal Type	Param.	Range of Value
All	SNR (dB)	$\mathcal{U}(-20, 20, 8)$
	f_{c} (MHz)	$\mathcal{U}(-f_s/2, f_s/2, 10)$
	ϕ (rad)	$\mathcal{U}(-\pi/4, \pi/4, \pi/180)$
Rect (and pulsed)	t_{pw} (µs)	$\mathcal{U}(1, 100, 1)$
	t_{prf} (kHz)	$\mathcal{U}(10, 50, 1)$
	N_{pulse}	$\mathcal{U}(2, 10, 1)$
LFM, FMCW	B_{chirp} (MHz)	$\mathcal{U}(10, 100, 1)$
	t_{chirp} (µs)	$\mathcal{U}(10, 100, 10)$
	N_{chirp}	$\mathcal{U}(1, 20, 1)$
Barker	N_{chip}	$\{5, 7, 11, 13\}$
Frank, Px, P1	N_{chip}	$\{4, 9, 16\}$
P2	N_{chip}	$\{4, 16\}$
P3, P4	N_{chip}	$\mathcal{U}(4, 16, 1)$
Zadoff-Chu	N_{chip}	$\{3, 5, 7, 9, 11, 13, 15\}$

B. Wideband Radar Detection Dataset

While NIST-CBRS serves as a valuable baseline dataset, it is limited by a narrow 10 MHz band with SNR settings confined to 10 to 20 dB. Critically, it only features at most a single radar instance per frame, which limits its applicability in modelling dense RF scenarios. We introduce RadDet, a novel wideband dataset for radar spectrum detection. RadDet builds upon [20] and introduces 11 radar classes, including 6 new LPI polyphase codes (P1, P2, P3, P4, Px, Zadoff-Chu) and a new wideband frequency-modulated continuous wave (FMCW), all coexisting across a 500 MHz band. The key parameters used in RadDet are summarised in Table I. Here, $\mathcal{U}(a, b, c)$ denotes uniform random sampling from a to b with resolution c, while $\{\cdot\}$ denotes uniform random sampling from a discrete set. N_i denotes the quantity of the parameter i.

Unlike existing RF classification datasets [7], [18], [19], RadDet provides long sequences, each consisting of 1 million complex baseband I/Q samples ($\vec{x}_{\text{i}} + j\vec{x}_{\text{q}}$). We generate a total of $40,000$ radar frames and distribute the dataset in three parts. The training set contains $20,000$ signals, while the validation

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI. Downloaded on August 06,2026 at 15:39:50 UTC from IEEE Xplore. Restrictions apply.



(a) NIST-CBRS (b) RadDet-1T (c) RadDet-9T

(d) NIST-CBRS (e) RadDet-1T (f) RadDet-9T

Fig. 2. Radar detection results at 10 dB SNR. Ground truth frames in (a)–(c) correspond to model predictions in (d)–(f), respectively. The confidence scores of each bounding box prediction are also shown in (d)–(f).

TABLE II
RADAR DETECTION DATASET VARIANTS

Dataset	**Size**	N_{pool}	**Dim.**	**Res.** (µs, MHz)
NIST-CBRS ($f_s = 10$ MHz)	S	60	128^2	624, 0.08
	M	30	256^2	313, 0.04
	L	15	512^2	157, 0.02
RadDet-9T/1T ($f_s = 500$ MHz)	S	75	128^2	15.6, 3.91
	M	19	256^2	7.81, 1.95
	L	4	512^2	3.91, 0.98

and test sets contain $14,000$ and $6,000$ signals, respectively. Additive white Gaussian noise (AWGN) is added to each signal to simulate varying SNR settings. We sample SNR from a uniform distribution to produce signal frames that fall within -20 and 20 dB at a resolution of 8 dB. This means that there are more than 6,500 unique signals generated per SNR. The data splits are carefully chosen to maintain a similar inter-class distribution to [28] across each SNR.

To investigate wideband spectrum detection in different scenarios, we provide RadDet in two different radar environments: dense and sparse. Similar to NIST-CBRS, our sparse dataset (RadDet-1T) provides at most a single radar instance per frame whereby the probability of a radar being present in a scene is 50%. Our dense dataset (RadDet-9T) contains up to 9 radar instances per frame where the probability of background (noise-only) frames is 10%. RadDet-9T represents a conservative dense maritime radar environment, as described in [1], where up to 200 radars may be observed within a typical 2-second surveillance period. We consider uniform random sampling to independently generate different permutations of radar encounters and signal attributes for each frame. This means that radar activities in each frame are uniformly distributed across a fixed time horizon (2 ms) and a fixed frequency band (500 MHz). To ensure that the signal parameters chosen for each radar class reflect typical radars encountered in EW, we adopt the emitter parameter library from [32] to bound our sampling range. For example, we consider large sweep bandwidths of up to 100 MHz for LFM signals, whereas the bandwidths of LFM radars are typically much lower outside the context of EW [26]. We transform the baseband I/Q signals into spectrograms using the STFT. The approach in [29] is adapted to generate max-hold spectrograms as a compute-efficient way to represent wideband signals. Due to the large spread of signal parameters considered in RadDet, no window functions are applied to avoid introducing pre-processing bias towards certain classes. The size of each max-hold spectrogram is calculated by

$$\dim(t) = \frac{N - N_{\text{overlap}}}{N_{\text{pool}}(N_{\text{seg}} - N_{\text{overlap}})}, \tag{4}$$

$$\dim(f) = N_{\text{nfft}}, \tag{5}$$

where N_{nfft} denotes the number of frequency bins computed by the STFT, N_{seg} denotes length of the time segment used for computing the STFT, N_{overlap} denotes the overlap between consecutive time segments, and N_{pool} represents the number of time bins used in the max-hold operation which controls the level of compression applied. While the max-hold operation reduces temporal resolution, it is necessary to enable efficient processing of extremely long I/Q sequences. The degree of compression applied should be scenario-dependent in an EW context, and balance the trade-off between detection accuracy and speed. We provide the RadDet dataset as spectrograms in three resolutions as shown in Table II where bbox annotations are computed per (1) and (2). An example of resultant annotations is depicted in Fig. 1. Annotations are normalised and structured using the standard YOLO format [33].

IV. EXPERIMENTS

A. Detection Models

We establish a performance baseline for wideband radar spectrum detection by considering real-time detection models, all targeting multi-class radar detection on both the NIST-CBRS and RadDet datasets. We consider real-time object detectors that are competitive in computer vision tasks, including YOLOv3 [34], YOLOv6 [35], YOLOv9 [36], and RT-DETR [37]. These detectors provide a comprehensive representation of baseline performance, allowing us to investigate the trade-off between detection performance and inference speed across different input resolutions and radar environments.

We train and evaluate each model on a single Nvidia Tesla A100 GPU. The models are trained for 100 epochs using the AdamW optimiser with a linear learning rate scheduler, where the learning rate is bounded between 0.01 and 0.001, and a warm-up period of 3 epochs at 0.1. To improve generalisation and training stability, we normalise spectrogram inputs to the range $[0, 1]$ and consider contemporary data augmentation techniques, including translations (10%), scaling (50%), erasing (20%), mixup (20%), mosaic (2×2), and copy-paste (20%) [38]. We also train models without data augmentation to serve as a comparison. All models are evaluated on test

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI. Downloaded on August 06,2026 at 15:39:50 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PERFORMANCE OF DETECTION MODELS ON NIST-CBRS AND RADDET-9T/1T TEST SETS

Model	Size	#Param. (M)	NIST-CBRS (%)		RadDet-9T (%)		RadDet-1T (%)		FPS_{avg}
			mAP_{50}	$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	
YOLOv3	S	103.67	87.20	67.28	25.68	19.10	19.62	13.28	1023.0
YOLOv6-M		51.98	86.57	66.37	23.02	16.49	10.41	4.93	**1301.8**
YOLOv9-M		20.02	85.88	67.72	**25.93**	**19.47**	**20.10**	**13.29**	896.6
RT-DETR-L		31.99	**91.27**	**73.29**	24.70	17.77	18.75	12.52	228.5
YOLOv3	M	103.67	93.52	72.96	**35.69**	**29.82**	24.81	18.67	688.2
YOLOv6-M		51.98	92.97	75.86	31.45	25.96	22.68	16.49	**921.3**
YOLOv9-M		20.02	93.81	**79.12**	35.10	29.14	**26.03**	**19.73**	579.2
RT-DETR-L		31.99	**94.69**	78.70	26.20	19.43	20.07	14.55	210.5
YOLOv3	L	103.67	93.70	72.62	**60.37**	**53.97**	**31.85**	**25.41**	274.2
YOLOv6-M		51.98	88.23	70.82	43.57	38.14	24.66	18.18	**470.9**
YOLOv9-M		20.02	91.31	72.97	47.37	40.85	31.41	24.56	449.4
RT-DETR-L		31.99	**95.31**	**80.96**	29.34	19.90	20.90	15.25	177.7

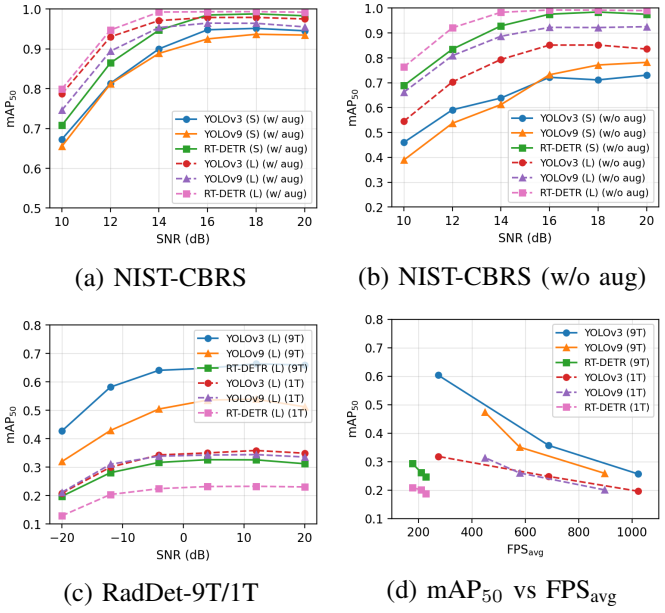
sets where the best models are selected based on their best validation performance during training.

B. Radar Detection Benchmark

We evaluate the baseline performance of detection models suitable for real-time applications on the NIST-CBRS and RadDet-9T/1T test sets through the following experiments: (i) comparison of baseline architectures; (ii) impact of time-frequency resolution on detection accuracy and speed; and (iii) changes in detection performance across different radar density environments and SNR settings. Performance is measured using inference speed, represented by the average frame rate (FPS_{avg}), and mean average precision (mAP) reported at two intersection-over-union (IoU) [39] settings: 50% (mAP_{50}) and 50 to 95% at 5% increments ($mAP_{50:95}$).

Table III presents a summary of results. The metrics mAP_{50} and $mAP_{50:95}$ presented in Table III are mean values computed across all 6 SNR intervals from the respective datasets. Several interesting observations can be made. First, data augmentation greatly enhances detection performance across all models without impacting inference speed, as shown in Fig. 3(a)–(b). Second, there is a distinct trade-off between mAP and FPS_{avg} across all models. Higher time-frequency resolution inputs provide better mAP but at the cost of lower FPS_{avg} due to increased computational demands. YOLOv3 at an input size of 128^2 (S) is 49% faster than at 256^2 (M), with a corresponding mAP_{50} reduction of 7% and 39% on NIST-CBRS and RadDet-9T, respectively. This trend is highlighted in Fig. 3(d). Third, the YOLO family of models outperforms RT-DETR on RadDet-9T, which is expected as the detection of small objects is a known limitation [37] of RT-DETR. Furthermore, RT-DETR relies on pre-trained embeddings from natural images, making it less suited for radar spectrum detection.

Lastly, the sparse radar environment in RadDet-1T poses challenges for all models across all SNR settings, as shown in Fig. 2. This challenge stems from two key factors: the presence of small objects (e.g., LPI radar classes) and the infrequent occurrence of radar pulses in a sparse signal environment. In contrast, radar classes in NIST-CBRS typically occupy larger bounding box footprints due to the lower spectrum bandwidth, providing a higher ratio of useful signal pixels to noise pixels for detection models, leading to better detection performance (Fig. 3). These characteristics of RadDet make it a challenging dataset for real-time detection models.



(a) NIST-CBRS (b) NIST-CBRS (w/o aug)

(c) RadDet-9T/1T (d) mAP_{50} vs FPS_{avg}

Fig. 3. Comparison of detection performance of models evaluated on NIST-CBRS and RadDet-9T/1T across different SNR settings. Unless otherwise noted, results reflect the use of data augmentation during training.

V. CONCLUSION

We have introduced RadDet, an open-source radar dataset designed for the challenging task of wideband radar spectrum detection. This dataset presents 11 classes of radar signals across 6 different SNR settings, 2 radar density environments, and 3 different time-frequency resolutions with corresponding time-frequency annotations. Our baseline results demonstrate the feasibility of applying real-time detectors for wideband radar spectrum detection, establishing a first-of-its-kind benchmark that explores the trade-off between time-frequency resolution, detection accuracy, and inference speed for real-time radar spectrum detection.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI. Downloaded on August 06,2026 at 15:39:50 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] Sue Robertson, *Practical ESM Analysis*, Artech House, 2019.

[2] Karen Haigh and Julia Andrusenko, *Cognitive Electronic Warfare: An Artificial Intelligence Approach*, Artech House, 2021.

[3] Bart Scheers, Djamel Teguig, and Vincent Le Nir, "Wideband Spectrum Sensing Technique Based on Goodness-of-Fit Testing," in *2015 International Conference on Military Communications and Information Systems (ICMCIS)*. IEEE, 2015, pp. 1–6.

[4] Alexander Schmidt, Christoph Rügheimer, Florian Particke, Thomas Mahr, Holger Appel, and Hans-Georg Kölle, "Kurtosis-Based Approach for Detection of Targets in Noise," in *2016 17th International Radar Symposium (IRS)*. IEEE, 2016, pp. 1–3.

[5] Timothy J. O'Shea, Johnathan Corgan, and T. Charles Clancy, "Convolutional Radio Modulation Recognition Networks," in *Engineering Applications of Neural Networks: 17th International Conference, EANN 2016, Aberdeen, UK, September 2-5, 2016, Proceedings*. 2016, pp. 213–226, Springer.

[6] Timothy J. O'Shea, Johnathan Corgan, and T. Charles Clancy, "Unsupervised Representation Learning of Structured Radio Communication Signals," in *2016 First International Workshop on Sensing, Processing and Learning for Intelligent Machines (SPLINE)*. 2016, pp. 1–5, IEEE.

[7] Timothy J. O'Shea, Tamoghna Roy, and T. Charles Clancy, "Over-the-Air Deep Learning Based Radio Signal Classification," *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, no. 1, pp. 168–179, 2018, IEEE.

[8] Andres Vila, Donna Branchevsky, Kyle Logue, Sebastian Olsen, Esteban Valles, Darren Semmen, Alex Utter, and Eugene Grayver, "Deep and Ensemble Learning to Win the Army RCO AI Signal Classification Challenge," in *Proceedings of the 18th Python in Science Conference*, 2019, pp. 21–26.

[9] Kyle Logue, Esteban Valles, Andres Vila, Alex Utter, Darren Semmen, Eugene Grayver, Sebastian Olsen, and Donna Branchevsky, "Expert RF Feature Extraction to Win the Army RCO AI Signal Classification Challenge," in *Proceedings of the 18th Python in Science Conference*, 2019, pp. 8–14.

[10] Thien Huynh-The, Quoc-Viet Pham, Toan-Van Nguyen, Thanh Thi Nguyen, Rukhsana Ruby, Ming Zeng, and Dong-Seong Kim, "Automatic Modulation Classification: A Deep Architecture Survey," *IEEE Access*, vol. 9, pp. 142950–142971, 2021.

[11] Tim O'Shea, Tamoghna Roy, and T. Charles Clancy, "Learning Robust General Radio Signal Detection Using Computer Vision Methods," in *2017 51st Asilomar Conference on Signals, Systems, and Computers*. IEEE, 2017, pp. 829–832.

[12] Tao Chen, Shilian Zheng, Kunfeng Qiu, Luxin Zhang, Qi Xuan, and Xiaoniu Yang, "Augmenting Radio Signals With Wavelet Transform for Deep Learning-Based Modulation Recognition," *IEEE Transactions on Cognitive Communications and Networking*, 2024.

[13] Thien Huynh-The, Van-Phuc Hoang, Jae-Woo Kim, Minh-Thanh Le, and Ming Zeng, "WaveNet: Towards Waveform Classification in Integrated Radar-Communication Systems with Improved Accuracy and Reduced Complexity," *IEEE Internet of Things Journal*, 2024.

[14] Thien Huynh-The, Van-Sang Doan, Cam-Hao Hua, Quoc-Viet Pham, Toan-Van Nguyen, and Dong-Seong Kim, "Accurate LPI Radar Waveform Recognition With CWD-TFA for Deep Convolutional Network," *IEEE Wireless Communications Letters*, vol. 10, no. 8, pp. 1638–1642, 2021.

[15] Stefan Scholl, "Fourier, Gabor, Morlet, or Wigner: Comparison of Time-Frequency Transforms," *arXiv Preprint arXiv:2101.06707*, 2021.

[16] Ioanna Kakalou, Danai Papadopoulou, Theofanis Xifilidis, Kostas E. Psannis, Katherine Siakavara, and Yutaka Ishibashi, "A Survey on Spectrum Sensing Algorithms for Cognitive Radio Networks," in *2018 7th International Conference on Modern Circuits and Systems Technologies (MOCAST)*. IEEE, 2018, pp. 1–4.

[17] Yuan Gong, Yu-An Chung, and James Glass, "AST: Audio Spectrogram Transformer," July 2021, arXiv:2104.01778 [cs].

[18] Zi Huang, Akila Pemasiri, Simon Denman, Clinton Fookes, and Terrence Martin, "Multi-task Learning For Radar Signal Characterisation," in *2023 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW)*, 2023, pp. 1–5.

[19] Anu Jagannath and Jithin Jagannath, "Multi-task Learning Approach for Automatic Modulation and Wireless Signal Classification," in *ICC 2021-IEEE International Conference on Communications*. 2021, pp. 1–7, IEEE.

[20] Zi Huang, Akila Pemasiri, Simon Denman, Clinton Fookes, and Terrence Martin, "Multi-stage Learning for Radar Pulse Activity Segmentation," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 7340–7344.

[21] W. Max Lees, Adam Wunderlich, Peter J. Jeavons, Paul D. Hale, and Michael R. Souryal, "Deep Learning Classification of 3.5-GHz Band Spectrograms With Applications to Spectrum Sensing," *IEEE Transactions on Cognitive Communications and Networking*, vol. 5, no. 2, pp. 224–236, 2019.

[22] Akila Pemasiri, Zi Huang, Fraser Williams, Ethan Goan, Simon Denman, Terrence Martin, and Clinton Fookes, "Automatic Radar Signal Detection and FFT Estimation using Deep Learning," in *2024 17th International Conference on Signal Processing and Communication System (ICSPCS)*, 2024, pp. 1–5.

[23] Zhe Geng, He Yan, Jindong Zhang, and Daiyin Zhu, "Deep Learning for Radar: A Survey," *IEEE Access*, vol. 9, pp. 141800–141818, 2021.

[24] Jian Wan, Xin Yu, and Qiang Guo, "LPI Radar Waveform Recognition Based on CNN and TPOT," *Symmetry*, vol. 11, no. 5, pp. 725, 2019.

[25] Victoria Clerico, Jorge González-López, Gady Agam, and Jesús Grajal, "LSTM Framework for Classification of Radar and Communications Signals," in *2023 IEEE Radar Conference (RadarConf23)*. IEEE, 2023, pp. 1–6.

[26] Phillip E Pace, *Detecting and Classifying Low Probability of Intercept Radar*, Artech House, 2009.

[27] Anu Jagannath and Jithin Jagannath, "Multi-task Learning Approach for Modulation and Wireless Signal Classification for 5G and Beyond: Edge Deployment via Model Compression," June 2022, arXiv:2203.00517 [cs, eess].

[28] Raied Caromi, Michael Souryal, and Timothy A. Hall, "RF Dataset of Incumbent Radar Signals in the 3.5 GHz CBRS Band," *Journal of Research of the National Institute of Standards and Technology*, vol. 124, pp. 1, 2019.

[29] Raied Caromi, Alex Lackpour, Kassem Kallas, Thao Nguyen, and Michael Souryal, "Deep Learning for Radar Signal Detection in the 3.5 GHz CBRS Band," in *2021 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*. IEEE, 2021, pp. 1–8.

[30] Sanjoy Basak, Sreeraj Rajendran, Sofie Pollin, and Bart Scheers, "Combined RF-Based Drone Detection and Classification," *IEEE Transactions on Cognitive Communications and Networking*, vol. 8, no. 1, pp. 111–120, 2021.

[31] Nasim Soltani, Vini Chaudhary, Debashri Roy, and Kaushik Chowdhury, "Finding Waldo in the CBRS Band: Signal Detection and Localization in the 3.5 GHz Spectrum," in *GLOBECOM 2022-2022 IEEE Global Communications Conference*. IEEE, 2022, pp. 4570–4575.

[32] David. Lynch Jr., *An Introduction to RF Stealth.*, Radar, Sonar and Navigation. Institution of Engineering & Technology, Stevenage, 2nd ed. edition, 2021.

[33] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 779–788.

[34] Joseph Redmon and Ali Farhadi, "YOLOv3: An Incremental Improvement," *arXiv Preprint arXiv:1804.02767*, 2018.

[35] Chuyi Li, Lulu Li, Hongliang Jiang, Kaiheng Weng, Yifei Geng, Liang Li, Zaidan Ke, Qingyuan Li, Meng Cheng, Weiqiang Nie, et al., "YOLOv6: A Single-Stage Object Detection Framework for Industrial Applications," *arXiv Preprint arXiv:2209.02976*, 2022.

[36] Chien-Yao Wang, I-Hau Yeh, and Hong-Yuan Mark Liao, "YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information," *arXiv Preprint arXiv:2402.13616*, 2024.

[37] Yian Zhao, Wenyu Lv, Shangliang Xu, Jinman Wei, Guanzhong Wang, Qingqing Dang, Yi Liu, and Jie Chen, "DETRs Beat YOLOs on Real-Time Object Detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16965–16974.

[38] Golnaz Ghiasi, Yin Cui, Aravind Srinivas, Rui Qian, Tsung-Yi Lin, Ekin D Cubuk, Quoc V Le, and Barret Zoph, "Simple Copy-Paste Is a Strong Data Augmentation Method for Instance Segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 2918–2928.

[39] Mark Everingham, Luc Van Gool, Christopher KI Williams, John Winn, and Andrew Zisserman, "The PASCAL Visual Object Classes (VOC) Challenge," *International Journal of Computer Vision*, vol. 88, pp. 303–338, 2010.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI. Downloaded on August 06,2026 at 15:39:50 UTC from IEEE Xplore. Restrictions apply.